

v5 Aegis Home System Draft 5

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to document:** v5 Aegis Home System Draft 5 V2.0

Revision Necessity — Three Drivers

- 1. **Empirical data superseded** — 2024-2025 peer-reviewed publications invalidate V1.0 material property values
- 2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) vs ~115 (forecast used in V1.0); design basis updated accordingly
- 3. **Standards/regulations updated** — One or more referenced codes superseded; V1.0 compliance citations rendered obsolete

V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Solar Cycle 25 risk framing	V1.0: Risk basis SSN ~115 (original SC25 forecast)	V2.0: Revised to SSN ~137 (NOAA SWPC Q1 2026 actuals); 10-year Carrington-class probability +20%	Observed solar activity exceeded forecast; design basis was non-conservative	NOAA SWPC Solar Cycle 25 Progression Report Q1 2026
BFRP rebar design standard	V1.0: ACI 440.1R-15	V2.0: ACI 440.11-25 (Seismic Type S); design tensile 1000 MPa	Seismic provisions absent in V1.0 — required for CA/PNW deployment; 11-year code update	ACI 440.11-25 January 2025; ACI 318-25
ZrB2-SiC sintering temperature	V1.0: SPS at 1900–2100°C, 30–50 MPa, 20 min	V2.0: Flash Sintering at 1800°C, 50 MPa, 5 min; >98% TD achieved	40% energy reduction; lower T preserves FSS layer integrity; equivalent density	Johnson et al., Acta Materialia 278 (2024) 120223

Parameter	V1.0	V2.0	Reason	Primary Source
Structural ceramic option	V1.0: Monolithic ZrB2-SiC only; diamond tooling required; brittle	V2.0: Ti3AlC2 MAX Phase hybrid added; carbide-machinable; KNE quasi-ductile	35% machining cost reduction; eliminates brittle shatter at stress concentrations	Barsoum MAX Phases; Anasori & Gogotsi 2023
Concrete binder	V1.0: Portland cement OPC; ~505 kg CO2/m3; 40 MPa	V2.0: Geopolymer binder; ~113 kg CO2/m3; 80 MPa (RILEM 2025)	78% CO2 reduction; 2x strength; ACI 318-25 geopolymer provisions now available	RILEM TC-281 2025; ACI 318-25; ASTM C1920-2024

Impact If V1.0 Retained (Criticality Matrix)

Parameter Changed	Consequence of Non-Upgrade	Severity
Solar Cycle 25 risk framing	Probability underestimated; design basis not conservative for observed SC25 activity	MEDIUM
BFRP rebar design standard	Non-compliant with SDC D/E/F seismic zones; limits geographic deployment	HIGH
ZrB2-SiC sintering temperature	Excess 400°C degrades embedded FSS; higher energy cost vs V2.0 process	MEDIUM
Structural ceramic option	Diamond tooling required; brittle fracture risk at fastener/joint stress concentrations	MEDIUM
Concrete binder	505 vs 113 kg CO2/m3; sustainability mandate unmet; lower strength ceiling	MEDIUM

Unchanged Elements

All design philosophy, fundamental equations (GIC induction, Joule heating, eddy current models), material selection rationale, product architecture, assembly methodology, and Phoenix Protocol structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | v5 Aegis Home System Draft 5 V1.0 archived; V2.0 is the active specification as of June 2026.